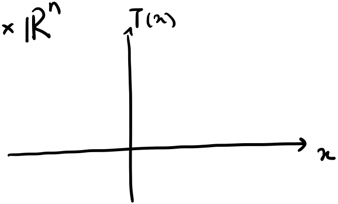
Operators, Graphs (Relations) and Generalized Equations

$T : \mathbb{R}^n \longrightarrow \mathbb{R}^n$ possibly multiple valued

(main example: ∇f, or ∂f)

Graph of T : Subset of $\mathbb{R}^n \times \mathbb{R}^n$

↳ $\{ (x, T(x)) : x \in \mathbb{R}^n \}$



Graph also called a relation R

Main example: $R = \{ (x, \partial f(x)) \}$.

Generalized Equations : $T(x) = 0$ for T single-valued

For multiple valued: $\boxed{0 \in T(x)}$

Also written for relations: $\boxed{0 \in R(x)}$

Relations: Common Operations

Inverse : R subset of $\mathbb{R}^n \times \mathbb{R}^n$

$$R^{-1} = \{(y,x) : (x,y) \in R\}$$

Composition: relations R, S

$$R \circ S = \{(x,y) : \exists z \text{ with } (x,z) \in S,\ (z,y) \in R\}$$

Scalar mult: $\alpha R = \{(x, \alpha y) : (x,y) \in R\}$

Addition: $R + S = \{(x, y+z) : (x,y) \in R,\ (x,z) \in S\}$

<u>Resolvent</u> : $S = (I + \lambda R)^{-1} = \{(x + \lambda y, x) : (x,y) \in R\}$

Monotone and Maximally Monotone Relations

A relation R is called <u>monotone</u> if

$$\langle u-v, x-y\rangle \geq 0 \quad \forall (x,u), (y,v) \in R$$

A monotone relation R is called <u>maximally monotone</u> if it is not contained in any monotone relation R'.

Primary (motivating) example

We have seen: ∂f is monotone <u>iff</u> f is convex.

∂f is maximally monotone if f is <u>closed</u> (and convex)

Resolvent and Cayley Operator

F monotone relation, $\lambda \geq 0$

Resolvent: $R = (I + \lambda F)^{-1}$: non-expansive

Cayley Operator: $\underline{C = 2R - I} = 2(I + \lambda F)^{-1} - I$

if $\lambda \geq 0$, F monotone $\Rightarrow$ C is non-expansive.

$\underline{R = \frac{1}{2} I + \frac{1}{2} C}$ $\Rightarrow$ R is averaged, in particular $\frac{1}{2}$-averaged, i.e., <u>firmly non-expansive</u>.

<u>Fixed points of R are fixed points of C.</u>

C is non-expansive, thus R firmly non-expansive 1/2

First we show: F monotone, $\lambda \geq 0 \Rightarrow R = (I + \lambda F)^{-1}$ <u>single-valued.</u>

Need to show: $(x,u), (y,v) \in R$, then $x = y \Rightarrow u = v$.

$$(u,x), (v,y) \in (I + \lambda F)$$

$$\Leftrightarrow \quad x \in u + \lambda F(u), \quad y \in v + \lambda F(v)$$

$$\Rightarrow \quad x - y \in u - v + \lambda (F(u) - F(v))$$

$$\Rightarrow \quad \langle u-v, x-y \rangle \in \|u-v\|_2^2 + \underbrace{\lambda \langle u-v, F(u) - F(v) \rangle}_{\geq 0}$$

$$\Rightarrow \quad \langle u-v, x-y \rangle \geq \|u-v\|_2^2$$

Thus: if $x = y \Rightarrow u = v$.

C is non-expansive, thus R firmly non-expansive 2/2

$C = 2R - I.$

$\boxed{\|C(x) - C(y)\| \le \|x-y\|}$ $\longleftarrow$ want to show.

$$\|C(x) - C(y)\| = \|(2u - x) - (2v - y)\|^2$$

$\left(\text{where } (x,u), (y,v) \in R \quad \text{i.e. } \begin{array}{l} u = R(x) \\ v = R(y) \end{array}\right)$

$$= \|2(u-v) - (x-y)\|_2^2$$

$$= \left(4\left(\|u-v\|_2^2 - \cancel{4}\langle u-v, x-y\rangle\right)\right) + \|x-y\|_2^2$$

$$\le \|x-y\|_2^2$$

$\Rightarrow$ C is non-expansive

and R is firmly non-expansive.

Next:

Gen Equ: $0 \in A(x) + B(x)$

Will show: $T = \underline{C_A \circ C_B}$

Fix T are the sol'ns to generalized equ.

<u>But</u> all we know is that C_A, C_B non-expansive

hence T is non-exp.

Not enough to show $z_{t+1} \leftarrow T(z_t)$ converges.

Instead: $(\underline{1}-\alpha)\underline{I} + \alpha T$

For $\alpha = 1/2$

This gives: Douglas - Rachford splitting.

In the dual this is ADMM.